

port_topology_aware_test

tests/regression/ port_topology_aware_test.sh — operator guide

Revision: 1 **Last modified:** 2026-07-01T00:00:00Z **Status:**
Active standing regression guard (§11.4.135) for BUGFIX-PORTS.

Companion (§11.4.18) to the in-source documentation block at the top of the script. Pairs with docs/research/existing_test_bluffs_audit/README.md.

Overview

A standing regression guard that proves tests/run-tests.sh never reports a **healthy, serving** proxy port as a **failure**. The pre-fix test_ports treated *any* port that was IN USE as FAIL, so the running proxy's own listening ports (squid 53128, dante 51080) were reported as failures and the whole suite exited 1 against a perfectly healthy serving proxy — a §11.4.1 false-FAIL (as forbidden as a false-PASS).

It does **not** grep the source. It exercises the **real** decision code: it extracts the current pure port_verdict() from tests/run-tests.sh and drives the full §11.4.3 topology truth-table, asserting the healthy “owner serving + port listening” case returns PASS (defect ABSENT). Because it tests the **pure** decision function, it is fully deterministic and never depends on live podman/ss state nor touches any container or the data plane.

Prerequisites

- bash (the extracted port_verdict uses bash local; the probe runs under set -euo pipefail).
- awk, mktemp. No network, no container access, no privileges.

Usage examples

```
# GREEN guard (default) – assert the real port_verdict truth-table
is honest:
tests/regression/port_topology_aware_test.sh
    # exit 0 = PASS

# RED reproduce – replicate the pre-fix logic and assert the
bluff:
RED_MODE=1 tests/regression/port_topology_aware_test.sh
    # exit 0 = defect reproduced

# Runs automatically inside the suite:
bash tests/run-tests.sh #
    test_regression_guards()
```

§11.4.115 RED_MODE polarity

RED_MODE	What it runs	PASS means
0 (default)	the REAL port_verdict() from tests/run-tests.sh	the healthy serving port (owner_serving=yes, listening=yes) → PASS, plus FAIL/PASS/ SKIP cells correct → the fix is present (GREEN guard)
1	the PRE-FIX replica (listening => FAIL)	the healthy serving port is classified FAIL → the bluff reproduces

A RED_MODE=1 run that *cannot* reproduce (replica does not FAIL the healthy port) is a finding per §11.4.7, not a pass.

The honest truth-table the GREEN guard asserts:

owner_serving	listening	verdict	meaning
yes	yes	PASS	service up and serving (squid/dante healthy)
yes	no	FAIL	owner up + publishing but nothing listening
no	no	PASS	

owner_serving	listening	verdict	meaning
			pre-start: port free, ready for ./start
no	yes	SKIP	pre-start, but a non-project process holds the port — readiness not assertable (§11.4.3)

Edge cases

- **Fix reverted** (e.g. port_verdict flipped so a serving port returns FAIL) → the GREEN guard reports REGRESSION: ... HEALTHY=FAIL ... and exits 1. This is the §1.1 paired-mutation behaviour: flipping the PASS/FAIL branch makes the guard FAIL (proven byte-identical-restore in BUGFIX-PORTS).
- **58080 held by a foreign host process** → in the live suite the control-API port has no project owner publishing it; if a non-project process occupies it the per-port check is a SKIP (not a false-FAIL blaming the proxy, not a false-PASS). Both occupied→SKIP and free→PASS keep the suite at 0 failures — the guard tests the pure decision so it is unaffected by that live variation.
- **podman/ss absent at suite runtime** → _ports_check_one resolves owner_serving=no and classifies on listener presence; the pure guard does not depend on either tool.

Internal behaviour

- `#!/bin/sh, set -eu`; POSIX-only constructs in the outer script (the bash-only port_verdict runs inside an explicit bash probe — §11.4.67).
- Extraction: `awk '/^port_verdict\(\) \/{f=1} f{print} /\)/{if(f){exit}}'` pulls the live function (closing } at column 0).
- Writes one evidence file per run under qa-results/regression/portstopology/ (gitignored).

Related

- Fix site: tests/run-tests.sh — port_verdict() + test_ports() / _ports_check_one() (§11.4.3 topology-aware dispatch).
- docs/research/existing_test_bluffs_audit/README.md — the bluff audit (port false-FAIL + B7 SKIP-as-PASS).

- `tests/run-tests.sh` — registers this guard via `test_regression_guards()`.

Last verified

2026-07-01 — `sh -n + bash -n parse-clean`; RED reproduces (VERDICT=FAIL for a healthy serving port); GREEN proves the fix (HEALTHY=PASS NOTSERVING=FAIL PRESTART_FREE=PASS PRESTART_BUSY=SKIP); §1.1 mutation (flip the PASS/FAIL branch) makes the GREEN guard FAIL with `bash -n` still clean (assertion, not parse error), and `tests/run-tests.sh` restores byte-identical (md5 2f4d3e4bf33cd391036cee595839d439). Full suite against the healthy running proxy: 43 run / 37 pass / 6 skip / 0 fail, exit 0.